Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_33

1. Summary of Key Deviations

The analysis of Run_33 reveals a healthy system with no significant faults. All observed deviations are minor and consistent with nominal operational variance or very slight calibration drifts.

- **Flow_Bypass (Minor):** A minor positive **offset** is observed, peaking at **+8.4%**.
- **Flow_Pump (Minor):** A corresponding minor positive **offset** of around **+5%** is seen on the pump flow.
- **p_Combustion (Minor):** A very slight positive **offset** of about **+4-6%** is present.
- **Flow_Engine1-4 (Minor):** All engine flows show a negligible positive **offset** of around **+4%** only during the high-throttle phase.
- **All Other Signals (Nominal):** All other pressure and flow signals show no significant deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**No Fault**	**Nominal Operation**	All signals are within or very close to their expected operational range. The minor positive offsets on flow and pressure are slight and could represent normal system variance, minor sensor calibration drift, or a system that is running slightly "hot" but not in a fault condition. There are no critical, major, or physically contradictory deviations.
2	**Main Fuel Pump**	**Minor Over-performance**	The pump could be delivering ~5% more flow than commanded. This would explain the minor positive offsets on `Flow_Pump`, `Flow_Bypass`, and `p_Combustion`. This is the most likely "fault," but the deviation is so small it may be within acceptable tolerance.

3	**System-wide Sensor Calibration**	**Minor Positive Bias**	It's possible that several sensors (`Flow_Pump`, `Flow_Bypass`, `p_Combustion`) all share a slight positive calibration bias. This would explain why they are all slightly high without any actual physical anomaly.
4	**Fuel Metering Unit (FMU)**	**Minor Over-delivery**	A slight miscalibration in the FMU could cause it to deliver slightly more fuel than commanded, explaining the minor positive engine flow and combustion pressure. However, this is a very small deviation.
5	**Bypass Valve**	**Slightly High Cracking Pressure**	A slightly higher-than-normal cracking pressure on the bypass valve could lead to a small increase in system pressure and a corresponding small increase in bypass flow, but the observed deviations are minimal.

3. Most Probable Hypothesis

The most probable hypothesis is **No Fault**. The system is operating within its nominal parameters.

Reasoning: There are no significant fault signatures in this run. All observed deviations are minor (mostly below 10%) and are physically consistent with a healthy system that might be operating at the higher end of its normal performance tolerance. There are no dropouts, no saturations, no major offsets, and no contradictory readings. The slight positive offsets in flow and pressure are not indicative of a broken component but rather of a system that is functioning correctly, with only very small, acceptable variations from the "perfectly healthy" baseline.